a process in which natural selection comes up with the same set of adaptations to a particular way of life, reshaping body parts or whole animals until they outwardly look the same. Convergence is responsible for a whole series of striking similarities in the animal kingdom. It can make the task of tracing evolution extremely difficult, which is why the theories regarding animal classification often change.

EURASIAN MOLE

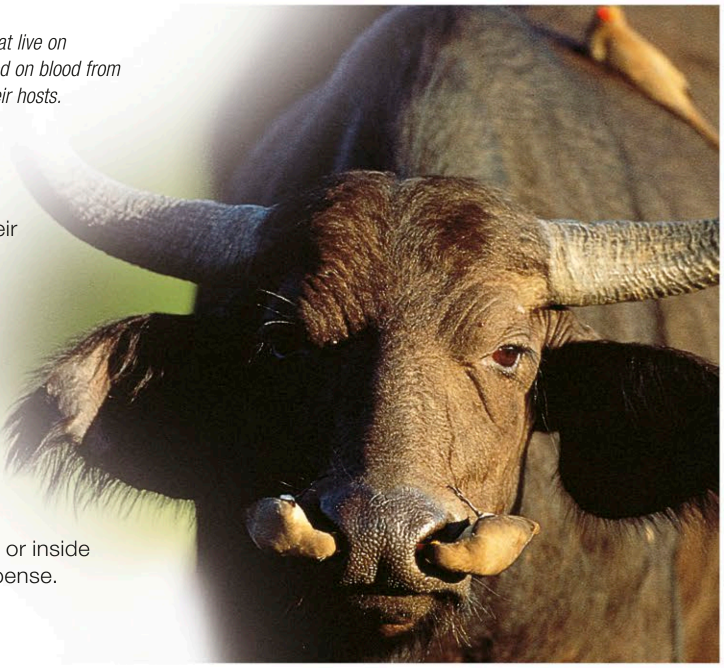
MARSUPIAL MOLE

SIMILAR SHAPES
Apart from their difference in color, Eurasian moles and marsupial moles are alike in many ways. However, they are not at all closely related; their similarities are a result of convergence.

Animal partnerships

Over millions of years, animals have evolved complex partnerships with each other and with other forms of life. In one common form of partnership, called mutualism, both species benefit from the arrangement. Examples are oxpeckers and large mammals, and corals and microscopic algae. Many partnerships are loose ones but, as with pollinating insects and plants, some are so highly evolved that the 2 partners cannot survive without each other. By contrast, a commensal partnership—such as that between remoras and their host fish (see p.522)—is one in which one species gains but the other neither gains nor loses.

Partnerships may appear to be mutually obliging, but each partner is driven purely by self-interest. If one partner can tip the balance in its favor, natural selection will lead it to do so. The ultimate outcome is parasitism, in which one animal, the parasite, lives on or inside another, entirely at the host's expense.



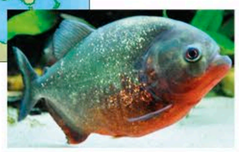
MIXED BLESSING
Oxpeckers eat ticks and other parasites that live on the skin of large animals, but they also feed on blood from wounds—a habit that is less helpful to their hosts.

Biogeography

The present-day distribution of animals is the combined result of many factors. Among them are continental drift and volcanic activity, which constantly reshape the surface of the earth. By splitting up groups of animals, and creating completely new habitats, these geological processes have had a profound impact on animal life. One of the most important effects can be seen on remote islands, such as Australia and Madagascar, which have been isolated from the rest of the world for millions of years. Until humans arrived, their land-based animals lived in total seclusion, unaffected by competitors from outside. The result is a whole range of indigenous species, such as kangaroos and lemurs, which are found nowhere outside their native homes.

Animals are separated when continents drift apart, and they are brought together when they collide. The distribution of animals is evidence of such events long after they occur. For example, Australasia and Southeast Asia became close neighbors long ago, but their wildlife remains entirely different: it is divided by "Wallace's line," an invisible boundary that indicates where the continents came together.

YELLOW-TAILED AFRICAN TETRA



SOUTH AMERICAN PIRANHA

AFRICAN CHARACINS

SOUTH AMERICAN CHARACINS

SPLIT BY CONTINENTAL DRIFT
The characin family—tropical freshwater fishes including tetras and piranhas—evolved where South America and Africa were once joined and so are now split across the two continents.

ARTIFICIAL SELECTION

The variations that natural selection works on are often difficult to see. One herring, for example, looks very much like another, while starlings in a flock are almost impossible to tell apart. This is because natural selection operates on a huge range of features among many individuals in a species. However, when animals are bred in controlled conditions, their hidden variations are very easy to bring out. Animal breeders rigorously concentrate on the reproduction of particular features, such as a specific size or color, and by selecting only those animals with the desired features they can exaggerate those features with remarkable speed. This process, called artificial selection, is responsible for all the world's domesticated animals and all cultivated plants.

CHIHUAHUA

GRAY WOLF

FAMILY LIKENESS
All domesticated dogs are descended from the gray wolf. Through artificial selection, individual breeds can be established in a very short time.

PERMIAN	TRIASSIC	JURASSIC	CRETACEOUS	PALEOGENE	NEOGENE	QUARTERNARY
Reptiles became the dominant land animals. The continents formed a single landmass. The period ended with the mass extinction—probably from climate change and volcanic activity—of over 75 percent of land species and over 90 percent of marine species.	The Triassic period marks the beginning of the "Age of Dinosaurs," when reptiles dominated life on Earth. They included flying pterosaurs, swimming forms such as nothosaurs and ichthyosaurs, and the first true dinosaurs. Early mammals also existed, but with reptiles in the ascendant they made up only a minor part of Earth's land-based fauna.	During this period, reptiles strengthened their position as the dominant form of animal life. They included a wide range of plant- and meat-eating species. There was a rich variety of vegetation and, toward the end of the Jurassic, flowering plants appeared, creating new opportunities for animals—particularly pollinating insects. Birds gradually evolved from dinosaurs. The first known bird, *Archaeopteryx*, probably took to the air over 150 million years ago.	The Cretaceous period saw rapid evolution in flowering plants and in animals that could use them as food. But the most striking feature of this period was the rise of the most gigantic land animals that have ever lived. They were all dinosaurs, and they included immense plant-eating sauropods, perhaps weighing up to 88 tons (80 tonnes), as well as colossal 2-legged carnivores, such as *Tyrannosaurus*, and its even bigger relative, *Carcharodontosaurus*. Toward the end of the Cretaceous, dinosaurs were already in decline, but they were wiped out by a mass extinction that also erased many other forms of life.	As the Earth cooled and dried, rainforests retreated to the equator and mammals diversified to take over from victims of the Cretaceous extinction event.	Grasslands replaced forests in many continental interiors. North and South America joined, disrupting warm ocean currents and leading to the formation of polar ice caps.	This period has seen abrupt changes in climate. Mammals have become supreme, and humans have become the dominant form of life.
251	200	145	65	23	2.6	PRESENT
	MESOZOIC ERA			CENOZOIC ERA		